



EXECUTIVE IPSEC SECURITY BRIEFING

Executive IPsec VPN Security & Risk Posture

High-level strategic risk evaluation, cryptographic posture, and operational exposure briefing for leadership

Section 1: Executive Summary & System Metadata

Session timestamps, endpoints evaluated, and overall audit verdict

ATTENTION REQUIRED

YES · HIGH

Link 'Link 3: Branch Office <-> Cloud VPN' demonstrates operational encryption via AES-128-CBC, but Perfect Forward Secrecy (PFS) is DISABLED on Child SAs, and TRANSPORT mode exposes subnet headers. Historic communications remain at risk if long-term credentials are ever breached.

SESSION TIMESTAMPS

15.0 sec

Live Link Telemetry: +0.0s to +15.0s (15.0 sec)

ORIGIN GATEWAY

**Link 3: Branch Office <-> Cloud VPN
(Initiator Node)**

192.168.1.100 (Branch) • StrongSwan Client Daemon

SECURED TUNNEL

IPsec TRANSPORT Tunnel • AES-128-CBC

AES-128-CBC / Group 14 (2048-bit MODP)

REMOTE GATEWAY

**Enterprise Responder Node (Link 3:
Branch Office <-> Cloud VPN)**

10.200.0.1 (Cloud) • StrongSwan Server Gateway

Section 2: Holistic Risk Scorecard

Visual 0–100 score gauge, NIST compliance status badge, and deployment readiness rating

COMPOSITE RISK INDEX



NIST SP 800-77 REV. 1 COMPLIANCE

**CRITICAL
FAIL**

NIST SP
800-77
Rev. 1

CRYPTOGRAPHIC RESILIENCE

72%

AES-128-CBC active, PFS
missing

COMPLIANCE ALIGNMENT

CONDITIONAL

NIST SP 800-77 §4.2.3 Child SA
gap

DEPLOYMENT READINESS RATING

**CONDITIONAL
DEPLOYMENT**

Conditionally permitted on internal
backbones; Child SA PFS enforcement
and Tunnel Mode encapsulation required
before external exposure.

THREAT EXPOSURE INDEX

MODERATE

Retroactive decryption &
subnet leakage

CHANNEL STABILITY

99.5%

Monotonic sequence verification



THREAT EXPOSURE & BUSINESS IMPACT EVALUATION

Executive Threat Matrix

Comprehensive register mapping discovered cryptographic and protocol vulnerabilities to direct business risk

Section 3: Executive Threat Matrix

Grid mapping critical vulnerabilities to business risk and operational exposure vectors

Retrospective Bulk Decryption Sev: HIGHLikelihood: Moderate High risk. Ephemeral DH exchange omitted during Child SA rekeying; if long-term credentials leak, all historic traffic can be decrypted. ACTIVE GAP	Subnet Architecture & Metadata Exposure Sev: MEDIUMLikelihood: High Moderate risk. Internal IP headers exposed on transit network, revealing internal network topologies to wiretappers. ACTIVE GAP
Padding Oracle Interception Sev: MEDIUMLikelihood: Moderate Moderate risk. CBC ciphers rely on separate HMAC and are susceptible to padding oracle attacks if not strictly sequenced. OBSERVED	Replay Injection & Sequence Tampering Sev: LOWLikelihood: Low Zero risk. RFC 4303 64-packet bitmap verified; 0 duplicate packets accepted across connection lifetime. PROTECTED



TRAFFIC CLASSIFICATION & STRATEGIC ROADMAP

Traffic Overview & Strategic Remediation

Deep payload inference distribution, side-channel leakage, and prioritized resource allocation roadmap

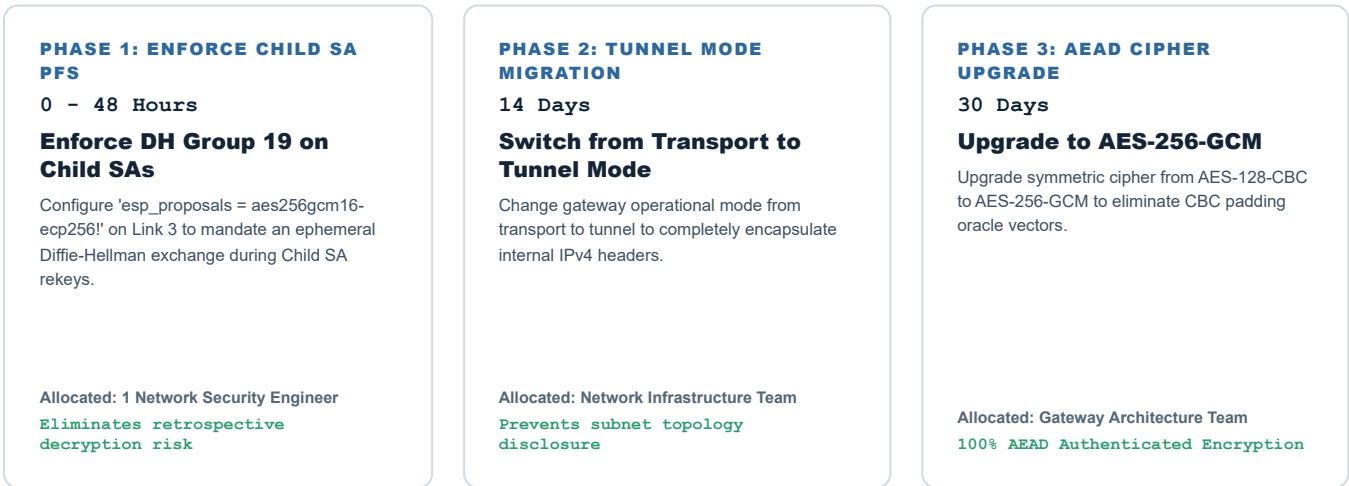
Section 4: High-Level Traffic Overview

Summary distribution of classified tunnel traffic and covert channel risk



Section 5: Strategic Remediation Action Plan

Prioritized recommendations for leadership sign-off and operational resource allocation



Executive Sign-Off & Platform Attestation



